

Critical behavior of a two-lattice model of antiferromagnetic phase transitions

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A two-lattice model of antiferromagnetic phase transitions is discussed in detail, using the Gell-Mann-Low formulation of renormalization-group methods and Wilson's ϵ expansion. Results of a previous work where the model was considered at the critical point are extended to deal with the situation in which the system is near but not at the critical temperature. A discussion is given of the behavior of various physical quantities and thermodynamic functions near the critical point, and of the equation of state. The equation of the critical line is also considered, and it is found that to order ϵ there are no corrections to the mean-field predictions of its critical behavior.

I. INTRODUCTION

In a previous paper¹ (hereafter referred to as I) we discussed the two-lattice model of antiferromagnetism in the context of Wilson's theory of critical phenomena.² In the long-wavelength approximation the theory can be recast in terms of a Lagrangian Euclidean field theory of two coupled scalar fields $\phi(x)$ and $\chi(x)$, whose vacuum expectation values represent the staggered and physical magnetizations, respectively. At the Néel temperature, the field $\phi(x)$ is massless but $\chi(x)$ has a large mass M of the order of the inverse of the lattice spacing. This Lagrangian field theory contains, in addition to the usual quartic self-couplings of the two fields, a mixing term of the form $g_3\phi^2\chi^2$.

The behavior of this system at the critical point was discussed in I by using the ϵ expansion, where $\epsilon = 4 - d$, d being the dimensionality of the lattice.² We found the rather obvious result that the scaling behavior of the massless field $\phi(x)$ is not affected by the presence of the massive field $\chi(x)$. The critical indices of a $2N$ -component antiferromagnetic system are therefore the same as those of a N -component ferromagnet.³ The purpose of this paper is to discuss more carefully the behavior of various thermodynamic functions near the critical point, and the corrections to scaling. Since scale invariance is only an asymptotic invariance near the critical point, there are well-known corrections to scaling, even if we neglect altogether the presence of the massive field. In this case they are identical to the ones obtained in the ferromagnetic case.³ We have here, in addition, corrections to scaling due to the coupling of the scaling field $\phi(x)$ to the massive field $\chi(x)$, and one of our purposes is to assess the relative importance of these corrections. They will be found to be consistently smaller than the first ones. Moreover, the antiferromagnetic critical point exhibits new features that are not present in the ferromagnetic case, such as, for example, the existence of a critical line that goes through the critical point in the phase

diagram of the system. It is also our purpose to discuss these new features by means of a careful analysis of the equation of state near the critical point.

In Sec. II we sketch the formalism used to discuss the critical behavior of our system, and in Sec. III we present and discuss our results.

II. GENERAL FORMALISM

In order to describe the behavior of the system under consideration near the critical point, we need to generalize the theory described in I. Our previous results were, strictly speaking, applicable only at the Néel point, because we worked with a massless field $\phi(x)$ and no external magnetic fields. We shall proceed step by step, by first reviewing the massless theory and then discussing the introduction of a temperature variable—or, equivalently, a mass for the field $\phi(x)$ —and the external magnetic fields.

A. Massless theory

In this case, the fields $\phi(x)$ and $\chi(x)$ have masses $m = 0$ and M , respectively. For reasons discussed at length in I, we find it convenient to scale the massive field χ to $M\chi$ from the start. This is done essentially because M goes to infinity at the antiferromagnetic critical point, and in this way the variation of M is absorbed into the coupling constants of the theory, since the propagator of the scaled field $\chi(x)$ is always of order one. The Lagrangian is then written as

$$\mathcal{L}(x) = \mathcal{L}_0(x) + \mathcal{L}_I(x), \quad (2.1)$$

where

$$\mathcal{L}_0(x) = -\frac{1}{2}(\vec{\nabla}\phi)^2 - (1/2M^2)(\vec{\nabla}\chi)^2 - \frac{1}{2}\chi^2 \quad (2.2)$$

and

$$\begin{aligned} \mathcal{L}_I(x) = & -\frac{1}{2} \frac{\delta M^2}{M} \frac{\chi^2(x)}{Z_\chi} - \frac{\delta m^2}{2} \frac{\phi^2(x)}{Z_\phi} - \frac{\bar{g}_1}{4!} Z_1 \frac{\phi^4}{Z_\phi^2} \\ & - \frac{\bar{g}_2 Z_2}{4!} \frac{\chi^4}{Z_\chi^2} - \frac{\bar{g}_3 Z_3}{4} \frac{\phi^2 \chi^2}{Z_\phi Z_\chi}. \end{aligned} \quad (2.3)$$

Here Z_ϕ , Z_χ are the wave-function renormalization constants, and Z_i ($i=1, \dots, 3$) are the coupling-constant renormalization constants. Moreover, $\bar{g}_i$ ($i=1, 2, 3$) are dimensional coupling constants in $d=4-\epsilon$ dimensions; g_2 and g_3 having dimensions even if $\epsilon=0$ because of our rescaling of the field $\chi(x)$. We call $\Gamma^{(N, N')}$ the one-particle irreducible (1PI) vertex functions of the theory with N , N' external lines of the fields ϕ , χ , respectively. Note that N , N' are always even. If we introduce an arbitrary normalization point μ^2 in momentum space, all constants in $\mathcal{L}_I(x)$ are determined by the normalization conditions

$$\Gamma^{(2,0)}(0) = \Gamma^{(0,2)}(-M^2) = 0, \quad (2.4)$$

$$\Gamma^{(2,0)}(\mu^2) = -\mu^2, \quad \Gamma^{(0,2)}(\mu^2) = -1 - \mu^2/M^2, \quad (2.5)$$

$$\Gamma^{(2,2)}|_{\text{SP}(\mu^2)} = -\bar{g}_3 = -g_3\mu^{\epsilon-2}, \quad (2.6)$$

$$\Gamma^{(4,0)}|_{\text{SP}(\mu^2)} = -\bar{g}_1 = -g_1\mu^\epsilon, \quad (2.7)$$

$$\Gamma^{(0,4)}|_{\text{SP}(\mu^2)} = -\bar{g}_2 = -g_2\mu^{\epsilon-4}, \quad (2.8)$$

where $\text{SP}(\mu^2)$ means the usual symmetric point for the four-point function.

The vertex functions of the theory satisfy the usual renormalization-group equations (RGE)

$$\left(\mu \frac{\partial}{\partial \mu} + \sum_{i=1}^3 \beta_i \frac{\partial}{\partial g_i} - N\gamma_\phi - N'\gamma_\chi \right) \times \Gamma^{(N, N')}(P_1, \dots, P_N; q_1, \dots, q_{N'}; \mu, M, g_i) = 0, \quad (2.9)$$

where the β and γ functions depend only on the dimensionless coupling constants g_i and on the ratio μ/M , and are determined by applying to Eq. (2.9) for the two- and four-point functions the normalization conditions as given by Eqs. (2.4)–

(2.8).⁴

In order to discuss the theory near but not at the critical point, we shall need 1PI vertex functions corresponding to any number of external lines of the fields ϕ , χ plus any number of soft-mass insertions of the fields ϕ^2 and χ^2 . However, because of the mixing implied by the term $g_3\phi^2\chi^2$ in the Lagrangian, these composite fields are not multiplicatively renormalizable. Therefore, we find it convenient to define the composite operators⁵

$$\Delta_1 = -\frac{1}{2} \left(\frac{Z_{11}}{Z_\phi} \phi^2 + \frac{Z_{12}}{Z_\chi} \chi^2 \right), \quad (2.10)$$

$$\Delta_2 = -\frac{1}{2} \left(\frac{Z_{21}}{Z_\phi} \phi^2 + \frac{Z_{22}}{Z_\chi} \chi^2 \right), \quad (2.11)$$

and define insertions in terms of them. A 1PI vertex function corresponding to N , N' , P , P' external lines of the fields ϕ , χ , Δ_1 , and Δ_2 , will be denoted by $\Delta_1^P \Delta_2^{P'} \Gamma^{(N, N')}$. We choose to determine the renormalization constants Z_{ij} ($i, j=1, 2$) in Eqs. (2.10) and (2.11) in perturbation theory by the following normalization conditions:

$$\begin{aligned} \Delta_1 \Gamma^{(2,0)}(k'; p_1, p_2) |_{3/4k^2=p_1^2=p_2^2=\mu^2} \\ = \Delta_2 \Gamma^{(0,2)}(k; q_1, q_2) |_{3/4k^2=q_1^2=q_2^2=\mu^2} = 1, \end{aligned} \quad (2.12)$$

$$\begin{aligned} \Delta_2 \Gamma^{(2,0)}(k; p_1, p_2) |_{3/4k^2=p_1^2=p_2^2=\mu^2} \\ = \Delta_1 \Gamma^{(0,2)}(k; q_1, q_2) |_{3/4k^2=q_1^2=q_2^2=\mu^2} = 0, \end{aligned} \quad (2.13)$$

which guarantees that $Z_{11}=Z_{22}=1$, $Z_{12}=Z_{21}=0$ in the free-field limit.

We shall also need in our future developments the RGE satisfied by the vertex function $\Delta_1^P \Gamma^{(N, N')}$, which takes the form

$$\begin{aligned} \left(\mu \frac{\partial}{\partial \mu} + \sum_{j=1}^3 \beta_j \frac{\partial}{\partial g_j} - N\gamma_\phi - N'\gamma_\chi - P\gamma_{11} \right) \Delta_1^P \Gamma^{(N, N')}(k_1, \dots, k_P; p, q; \mu, M, g_i) \\ = \gamma_{12} \sum_{s=1}^P \Delta_1^{P-1} \Delta_2 \Gamma^{(N, N')}(k_1, \dots, k_{s-1}, k_{s+1}, \dots, k_P; k_s; p, q, \mu, M, g_i), \end{aligned} \quad (2.14)$$

where we have only written explicitly the dependence of the vertex functions on the momenta of the external lines of the fields Δ_1 and Δ_2 . The β and γ functions in Eq. (2.14) are the same as those of Eq. (2.9). The remaining functions γ_{11} and γ_{12} are obtained by applying the normalization conditions given by Eqs. (2.12) and (2.13) to Eq. (2.14). In this way we obtain

$$\gamma_{11} = -2\gamma_\phi - \mu \frac{\partial}{\partial \mu} \Delta_1 \Gamma^{(2,0)} |_{3/4k^2=p_1^2=p_2^2=\mu^2}, \quad (2.15)$$

$$\gamma_{12} = -2\gamma_\chi - \mu \frac{\partial}{\partial \mu} \Delta_1 \Gamma^{(0,2)} |_{3/4k^2=p_1^2=p_2^2=\mu^2}. \quad (2.16)$$

B. Introduction of temperature

We want to discuss now the case in which the field $\phi(x)$ is not strictly massless but has a small nonvanishing mass m . However, instead of giving a mass directly to $\phi(x)$ we shall use here the so-called "soft quantization procedure,"^{3,6} which will allow us to use most of the results of the massless theory in performing actual calculations. The basic idea is to define a temperature parameter $t = T - T_N$ that vanishes at the critical point, and add to the Lagrangian defined in Eqs. (2.1)–(2.3) a term of the form

$$\Delta \mathcal{L}_t(x) = t \Delta_1 = -\frac{1}{2}t \left(\frac{Z_{11}}{Z_\phi} \phi^2 + \frac{Z_{12}}{Z_\chi} \chi^2 \right), \quad (2.17)$$

that is to say, a term proportional to the temperature times its conjugate field ϕ^2 , the energy density. The small mixing with the field χ^2 is required, as explained before, for reasons related to the renormalizability of the theory under consideration.

The 1PI vertex functions computed with the new Lagrangian $\mathcal{L} + \Delta \mathcal{L}_t$ will be denoted by $\bar{\Gamma}^{(N, N')}$ and

$$\left(\mu \frac{\partial}{\partial \mu} + \sum_{j=1}^3 \beta_j \frac{\partial}{\partial g_j} - N \gamma_\phi - N' \gamma_\chi - \gamma_t t \frac{\partial}{\partial t} \right) \bar{\Gamma}^{(N, N')}(p_i, q_j; t, M, \mu; g_i) = t \gamma_{12} \Delta_2 \bar{\Gamma}^{(N, N')}(0; p_i, q_j; t, M, \mu; g_i), \quad (2.19)$$

where $\gamma_t \equiv \gamma_{11}$ and $\Delta_2 \bar{\Gamma}^{(N, N')}$ is defined by performing a soft-mass insertion on each term of the right-hand side of Eq. (2.18).

C. Introduction of magnetic fields

In order to be able to discuss the equation of state, we must introduce the real and staggered magnetic fields $\mathcal{H}(x)$ and $\mathcal{H}_s(x)$, conjugate to the fields $\chi(x)$ and $\phi(x)$, respectively. They are the sum and the difference of the two magnetic fields $H_1(x)$ and $H_2(x)$ conjugate to the two sublattice magnetizations. We add therefore another term to the Lagrangian

$$\Delta \mathcal{L}_m(x) = \mathcal{H}(x) \frac{\chi(x)}{\sqrt{Z_\chi}} + \mathcal{H}_s(x) \frac{\phi(x)}{\sqrt{Z_\phi}}, \quad (2.20)$$

and define the total Lagrangian $\mathcal{L}_T(x)$

$$\mathcal{L}_T(x) = \mathcal{L}(x) + \Delta \mathcal{L}_t(x) + \Delta \mathcal{L}_m(x). \quad (2.21)$$

The partition function for the system is given by the path integral²

$$Z[\mathcal{H}, \mathcal{H}_s, t] = C(t) \int \mathcal{D}\phi \mathcal{D}\chi \exp \left(\int \mathcal{L}_T(x) d^d(x) \right), \quad (2.22)$$

where $C(t)$ is a normalization constant to be defined later on, and $\mathcal{D}\phi \mathcal{D}\chi$ is the measure of integration in the path integral. The generating functional for connected Green's functions—correlation functions in statistical mechanics—is defined as usual as

$$W[\mathcal{H}, \mathcal{H}_s, t] = \ln Z[\mathcal{H}, \mathcal{H}_s, t]. \quad (2.23)$$

This functional W is the Helmholtz free energy of the system. The real and staggered magnetizations $\mathfrak{M}(x)$ and $\mathfrak{M}_s(x)$, respectively, are then obtained as functional derivatives of W , as follows:

$$\mathfrak{M}(x) = \frac{\langle \chi(x) \rangle}{V} = \frac{1}{V} \frac{\delta W}{\delta \mathcal{H}(x)}, \quad (2.24)$$

$$\mathfrak{M}_s(x) = \frac{\langle \phi(x) \rangle}{V} = \frac{1}{V} \frac{\delta W}{\delta \mathcal{H}_s(x)}. \quad (2.25)$$

depend of course on the temperature t . They can be related to the vertex functions of the massless theory by the power-series expansion³

$$\begin{aligned} \Gamma^{(N, N')}(p_i, q_j; t, M, \mu; g_i) \\ = \sum_{P=0}^{\infty} \frac{t^P}{P!} \Delta_1^P \Gamma^{(N, N')}(0; p_i, q_j; M, \mu; g_i), \end{aligned} \quad (2.18)$$

from which one can derive, using Eq. (2.14), the RGE for $\bar{\Gamma}^{(N, N')}$. It takes the form

The Gibbs free energy per unit volume $\Gamma[\mathfrak{M}, \mathfrak{M}_s; t]$ can be defined by means of a functional Legendre transformation on W ,

$$\begin{aligned} \Gamma[\mathfrak{M}, \mathfrak{M}_s, t] &= (1/V) W[\mathcal{H}, \mathcal{H}_s, t] \\ &\quad - \frac{1}{V} \int d^d x [\mathcal{H}(x) \mathfrak{M}(x) + \mathcal{H}_s(x) \mathfrak{M}_s(x)], \end{aligned} \quad (2.26)$$

and from this definition of Γ one obtains the conjugate relations to Eqs. (2.24)–(2.25)

$$\mathcal{H}(x) = - \frac{\delta \Gamma}{\delta \mathfrak{M}(x)}, \quad (2.27)$$

$$\mathcal{H}_s(x) = - \frac{\delta \Gamma}{\delta \mathfrak{M}_s(x)}, \quad (2.28)$$

which, together with the renormalization group equation for the free energy Γ , will be the starting point of our discussion of the equation of state.

Note that $\mathfrak{M}(x)$ and $\mathfrak{M}_s(x)$ have different dimensions in mass units because of the rescaling we performed at the beginning of the massive field $\chi(x)$. Consequently, the two magnetic fields have also different dimensions.

The RGE for Γ can be obtained by observing that Γ is the generating functional for the 1PI vertex functions of the theory with no real and staggered magnetic fields. This simply means that $\Gamma[\mathfrak{M}, \mathfrak{M}_s, t]$ can be expanded as follows³:

$$\begin{aligned} \Gamma[\mathfrak{M}, \mathfrak{M}_s, t; M, \mu, g_i] &= (1/V) \ln C(t) \\ &\quad + \sum_{N, N'=0}^{\infty} \frac{\mathfrak{M}^{N'} \mathfrak{M}_s^N}{N'! N!} \bar{\Gamma}^{(N, N')} \\ &\quad (p_i = 0; q_j = 0; t, M, \mu; g_i). \end{aligned} \quad (2.29)$$

Let us now discuss the choice of the normalization factor $C(t)$ in the partition function defined by Eq. (2.22). In quantum field theory $C(t)$ is chosen by demanding that the vacuum-to-vacuum ampli-

tude—the analog of the partition function—be one at zero external fields. This choice means that the term $V^{-1} \ln C(t)$ in Eq. (2.29) exactly cancels the ultraviolet divergent term corresponding to $N = N' = 0$ in the expansion of $\Gamma[\mathfrak{M}, \mathfrak{N}, t]$. This procedure is very convenient in quantum field theory because one is always interested in computing vertex functions other than the vacuum amplitude. However, in statistical mechanics this choice means that the Helmholtz free energy is independent of temperature at zero external fields. We have, therefore, adopted a different choice for $C(t)$ by normalizing the energy correlation function by the condition

$$\Delta_1^2 \Gamma(k, -k; \mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i) \big|_{k^2 = \mu^2 = 0}, \quad (2.30)$$

where $\Delta_1^2 \Gamma$ is defined by performing n Δ_1 insertions in all the terms of the right-hand side of Eq. (2.29). Since one can easily show that

$$\Delta_1^2 \Gamma(0, 0; \mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i) = \frac{\partial^2 \Gamma}{\partial t^2}, \quad (2.31)$$

it is clear that Eq. (2.30) fixes $\ln C(t)$ up to a linear function of t . Therefore, the free energy and the entropy are defined with our normalization up to an additive constant, but the specific heat is fully determined.

From the expansion (2.29) and the RGE for $\Gamma^{(N, N')}$ given by Eq. (2.19) we obtain the RGE for the free energy

$$\left(\mu \frac{\partial}{\partial \mu} + \sum_{j=1}^3 \beta_j \frac{\partial}{\partial g_j} - \gamma_t t \frac{\partial}{\partial t} - \gamma_\phi \mathfrak{N} \frac{\partial}{\partial \mathfrak{N}} - \gamma_\chi \mathfrak{M} \frac{\partial}{\partial \mathfrak{M}} \right) \Gamma[\mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i] = t \gamma_{12} \Delta_2 \Gamma[\mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i], \quad (2.32)$$

where $\Delta_2 \Gamma$ is defined in the same way as $\Delta_1 \Gamma$. Strictly speaking, the derivatives with respect to $\mathfrak{N}$ and $\mathfrak{M}$ in Eq. (2.32) should be functional derivatives instead of ordinary ones. However, we shall restrict ourselves to the case of uniform external magnetic fields, in which case the magnetizations are independent of x , and the free energy Γ is just an ordinary function of them.

Using Euler's theorem for the free energy in $d = 4 - \epsilon$ dimensions, which is nothing but a statement about the dimensions of Γ in mass units,

$$\left[\mu \frac{\partial}{\partial \mu} + M \frac{\partial}{\partial M} + \left(1 - \frac{\epsilon}{2} \right) \mathfrak{N} \frac{\partial}{\partial \mathfrak{N}} + \left(2 - \frac{\epsilon}{2} \right) \mathfrak{M} \frac{\partial}{\partial \mathfrak{M}} + 2t \frac{\partial}{\partial t} + \epsilon - 4 \right] \Gamma = 0, \quad (2.33)$$

we can eliminate $\mu \partial/\partial \mu$ from Eq. (2.32), obtaining the final form for the RGE for the free energy. By differentiating this equation with respect to $\mathfrak{N}$ and $\mathfrak{M}$ and making use of Eqs. (2.27) and (2.28), we can then obtain the RGE for the magnetic fields $\mathcal{H}_s$ and $\mathcal{H}$, respectively. All these equations have the same general form:

$$\left[-M \frac{\partial}{\partial M} + \sum_{j=1}^3 \beta_j \frac{\partial}{\partial g_j} - \left(1 - \frac{\epsilon}{2} + \gamma_\phi \right) \mathfrak{N} \frac{\partial}{\partial \mathfrak{N}} - \left(2 - \frac{\epsilon}{2} + \gamma_\chi \right) \mathfrak{M} \frac{\partial}{\partial \mathfrak{M}} - (2 + \gamma_t t \frac{\partial}{\partial t} + d_F) \right] F = t \gamma_{12} \Delta_2 F, \quad (2.34)$$

where $F = \Gamma$, $\mathcal{H}$, or $\mathcal{H}_s$ and

$$\begin{aligned} d_F &= 4 - \epsilon, \\ d_{\mathcal{H}_s} &= 2 - \frac{1}{2}\epsilon - \gamma_\phi, \\ d_{\mathcal{H}} &= 3 - \frac{1}{2}\epsilon - \gamma_\chi. \end{aligned} \quad (2.35)$$

D. Solution of the renormalization-group equation

The final step is to solve the RGE (2.34). We know that the antiferromagnetic critical point corresponds to a fixed point of the renormalization group where the mass of the field $\phi(x)$ vanishes—so that the temperature t tends to zero—and the mass M of the field $\chi(x)$ goes to infinity.¹ Therefore, we perform a scale transformation of the form $M(\lambda) = M\lambda^{-1}$ and examine the behavior of the solution of Eq. (2.34) near $\lambda = 0$. This is done in the standard way, by defining effective coupling constants, magnetizations, and temperature that depend on λ and satisfy generalized Gell-Mann-Low equations of the form⁷

$$\lambda \frac{dg_i(\lambda)}{d\lambda} = \beta_i \left(g_j(\lambda), \frac{\mu}{M(\lambda)} \right), \quad i, j = 1, 2, 3 \quad (2.36)$$

$$\lambda \frac{d\mathfrak{N}(\lambda)}{d\lambda} = -\mathfrak{N}(\lambda) \left[1 - \frac{\epsilon}{2} + \gamma_\phi \left(g_i(\lambda), \frac{\mu}{M(\lambda)} \right) \right], \quad (2.37)$$

$$\lambda \frac{d\mathfrak{M}(\lambda)}{d\lambda} = -\mathfrak{M}(\lambda) \left[2 - \frac{\epsilon}{2} + \gamma_\chi \left(g_i(\lambda), \frac{\mu}{M(\lambda)} \right) \right], \quad (2.38)$$

$$\lambda \frac{dt(\lambda)}{d\lambda} = -t(\lambda) \left[2 + \gamma_t \left(g_i(\lambda), \frac{\mu}{M(\lambda)} \right) \right], \quad (2.39)$$

with the initial condition that these effective parameters take their given physical values g_i , $\mathfrak{N}$, $\mathfrak{M}$, t at $\lambda = 1$.

In these conditions, Eq. (2.34) reduces to an inhomogeneous first-order differential equation for the thermodynamic function F . This equation simply means that a scale transformation in M plus a change of all the couplings and additional pa-

rameters on which F depends along the "trajectories" defined by the Gell-Mann-Low equations (2.36)–(2.39), is equivalent to multiply F by d_F plus the effect of the inhomogeneous term that

vanishes at the critical point. The solution of this first-order differential equation relates $F(\lambda)$, the values of F when all parameters take their effective values $g_i(\lambda)$, $t(\lambda)$, etc., to $F(1)$ as follows⁸:

$$F(1) = F(\lambda) \exp\left(\int_1^\lambda d_F(\lambda') \frac{d\lambda'}{\lambda'}\right) - \int_1^\lambda \frac{d\lambda'}{\lambda'} t(\lambda') \gamma_{12}(\lambda') \Delta_2 F(\lambda') \exp\left(\int_1^{\lambda'} \frac{d\lambda''}{\lambda''} d_F(\lambda'')\right). \quad (2.40)$$

Let us now discuss the behaviour of the solutions of Eqs. (2.36)–(2.39) near the antiferromagnetic fixed point at $\lambda=0$. Because of the way in which we introduced the magnetic fields and the temperature in the problem, the β_i functions are the same as those of the massless theory discussed in I. Therefore, the effective coupling constants $g_i(\lambda)$ are also the same, and we can use the results of I, where we found that, as λ approaches zero, the effective coupling approach an infrared stable fixed point at $g_1 = g_1^* = O(\epsilon)$, $g_2 = g_3 = 0$, g_1^* being the standard fixed point for a one-component ferromagnetic system in $d = 4 - \epsilon$ dimensions.³ More precisely, the behavior of the effective coupling near $\lambda=0$ was found to be

$$\begin{aligned} g_1(\lambda) &\simeq g_1^* [1 + O(\lambda^\epsilon)] + O(\lambda^{2-\epsilon/2}), \\ g_2(\lambda) &\simeq b \lambda^{4-\epsilon}, \quad g_3(\lambda) \simeq c \lambda^{2-2\epsilon/3}, \end{aligned} \quad (2.41)$$

where b and c are some constants.

We now proceed to rewrite these results in a more convenient way for our purposes. Let us first consider the mixing constant g_3 . By examining the Feynman diagrams that, in perturbation theory, contribute to β_3 and γ_t , one can show that at the fixed point β_3 and γ_t are related by

$$\left. \frac{\partial \beta_3}{\partial g_3} \right|_{\text{FP}} = 2 - \epsilon - \gamma_t \Big|_{\text{FP}} = 4 - \epsilon - 1/\nu, \quad (2.42)$$

where we use the relation between the value of γ_t at the fixed point and the critical index ν of a one-component ferromagnetic system,³ because at the fixed point the massive field completely decouples. The normalization conditions given by Eqs. (2.4)–(2.8), (2.12), and (2.13), play a nontrivial role in the proof of Eq. (2.42), but, given the universality of the critical indices, the equation has a general validity. Since the remaining partial derivatives of β_3 vanish at the fixed point, we can immediately write down Eq. (2.34) for $g_3(\lambda)$ in the form

$$g_3\left(\lambda, \frac{\mu}{M}\right) = g_3 \lambda^{4-\epsilon-1/\nu} \frac{\xi_3(\lambda)}{\xi_3(1)}, \quad (2.43)$$

where

$$\xi_3(\lambda) = \exp\left[\int_0^\lambda \left(\frac{\beta_3}{g_3} - \frac{\beta_3}{g_3} \Big|_{\text{FP}}\right) \left(\frac{d\lambda'}{\lambda'}\right)\right]. \quad (2.44)$$

If we restrict ourselves to small values of λ —that is to say, if we solve the linearized Gell-Mann—

Low equations near the fixed point—the ratio $\xi_3(\lambda)/\xi_3(1)$ can be replaced by a constant. In the same way we can write

$$g_2\left(\lambda, \frac{\mu}{M}\right) = g_2 \lambda^{4-\epsilon} \frac{\xi_2(\lambda)}{\xi_2(1)}, \quad (2.45)$$

where $\xi_2(\lambda)$ is defined by an equation analogous to Eq. (2.44) with β_3 , g_3 replaced by β_2 , g_2 . Finally one can also show that $g_1(\lambda)$ can be written in the form

$$g_1(\lambda, \mu/M) = g(\lambda) + H(\lambda, \mu/M), \quad (2.46)$$

where $g(\lambda)$ is the effective coupling constant for the one-component ferromagnetic system³ and $H(\lambda, \mu/M)$ behaves, when $\lambda \rightarrow 0$, as

$$H(\lambda, \mu/M) \simeq \lambda^{6-\epsilon-2/\nu} [\text{const} + O(\lambda^2)]. \quad (2.47)$$

We can analyze in the same way the solutions of Eqs. (2.37)–(2.39) for the effective magnetization and temperatures. Defining the functions

$$\begin{aligned} \xi_l(\lambda) = \exp\left\{-\int_0^\lambda \left[\gamma_l\left(g_i(\lambda'), \frac{\mu}{M(\lambda')}\right) - \gamma_l(g_i^*, 0)\right] \frac{d\lambda'}{\lambda'}\right\}, \end{aligned} \quad (2.48)$$

where $l = \phi, \chi$, or t , the effective magnetizations and temperature can be written as

$$\mathfrak{M}\left(\lambda, \frac{\mu}{M}\right) = \mathfrak{M} \lambda^{1+\epsilon/2-\eta/2} \frac{\xi_\phi(\lambda)}{\xi_\phi(1)}, \quad (2.49)$$

$$\mathfrak{M}\left(\lambda, \frac{\mu}{M}\right) = \mathfrak{M} \lambda^{-2+\epsilon/2} \frac{\xi_\chi(\lambda)}{\xi_\chi(1)}, \quad (2.50)$$

$$t\left(\lambda, \frac{\mu}{M}\right) = t \lambda^{-1/\nu} \frac{\xi_t(\lambda)}{\xi_t(1)}, \quad (2.51)$$

where the critical index ν is the same as the one defined in Eq. (2.42), and the critical index η —the anomalous dimension of the field $\phi(x)$ —is defined by

$$\eta = 2\gamma_\phi \Big|_{\text{FP}}. \quad (2.52)$$

Moreover, in Eq. (2.48) use has been made of the fact that the field χ has no anomalous dimension, that is to say, γ_χ vanishes at the fixed point. We remind the reader once again that all the critical indices we introduced up to now are the same as those of a one-component ferromagnet.

III. DISCUSSION OF CRITICAL BEHAVIOR

We discuss in this section the critical behavior of various thermodynamic functions and physical quantities such as susceptibilities and correlation lengths. The starting point of the discussion is the solution (2.40) of the RGE for some thermodynamic function F (free energy, magnetic fields). This solution, that relates $F(\lambda)$ to $F(1)$, can be used at once to derive scaling laws near the critical point. We shall proceed in the same way as Di Castro, Jona-Lasinio, and Peliti,⁹ by eliminating the scale parameter λ from the solution in terms of the temperature t .

In order to eliminate λ one can use, for example, Eq. (2.51) for the effective temperature $t(\lambda)$. This equation states that, if we start from a given value t of the temperature for $\lambda=1$, after we make the scale transformation λ the temperature is mapped to the value $t(\lambda) > t$, since $\nu > 0$.

Now one can fix the *final* value of the temperature to some standard value $\bar{t}$ and define λ as the scale transformation needed in order to map t to $\bar{t}$. In this way, λ becomes a function of t , given by

$$\lambda = (t/\bar{t})^\nu. \quad (3.1)$$

In our calculations we have used

$$\bar{t} = A \mu^2 \xi_t(1), \quad (3.2)$$

where A is a dimensionless constant. Scaling laws for F are then obtained by neglecting the inhomogeneous term of Eq. (2.40) and using the fact that when λ is eliminated from Eq. (2.40) in the way defined by Eq. (3.1) the thermodynamic function F

depends effectively on one variable less. We shall not pursue this discussion here since this point is discussed at length for the ferromagnetic case in Ref. 9, and the results are very similar. We only mention the fact that whenever we needed the solution of the RGE for some thermodynamic function F in the calculations discussed below, we consistently kept track of contributions coming from the inhomogeneous term, which are some—but not all—of the sources of corrections to the scaling behavior. The remaining sources of correction to scaling are due to the fact that the effective parameters are not exactly at their fixed-point values. This has also been taken into account by using the solutions of the linearized Gell-Mann–Low equations near the fixed point.³

A. Loop-wise expansion

The scaling form of the equation of state near the critical point simply states that the scaled magnetic fields h and h_s —that is to say, the magnetic fields $\mathcal{H}$ and $\mathcal{H}_s$ divided by known powers of t —are two unknown functions of two variables, the scaled magnetizations. In order to obtain some further information, such as a more specific form of the equation of state, one needs, besides the solutions of the RGE, some independent estimates of the free energy Γ . Estimates of $\Delta_2 \Gamma$ are also needed in order to take into account the effects of the inhomogeneous term in the RGE for the free energy. Following standard practice we compute, Γ and $\Delta_2 \Gamma$ in the context of the loop-wise expansion.^{3,10} To the one-loop order, the free energy is given by

$$\Gamma[\mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i] = \frac{1}{V} S_t[\phi, \chi] \Big|_{\substack{\phi=\mathfrak{M} \\ \chi=\mathfrak{N}}} - \frac{1}{2} \int \frac{d^d q}{(2\pi)^d} \ln \det \tilde{G}(q) + \frac{1}{V} \ln C(t), \quad (3.3)$$

where the functional $S_t[\phi, \chi]$ is the total action defined in terms of the Lagrangian

$$\mathcal{L}_t(x) = \mathcal{L}(x) + \Delta \mathcal{L}_t(x), \quad (3.4)$$

and $\tilde{G}(q)$ is the inverse propagator matrix for the two coupled fields, given by

$$-\tilde{G}(q) = \begin{pmatrix} q^2 + t + \frac{1}{2} \bar{g}_1 \mathfrak{N}^2 + \frac{1}{2} \bar{g}_3 \mathfrak{M}^2 & \bar{g}_3 \mathfrak{N} \mathfrak{M} \\ \bar{g}_2 \mathfrak{N} \mathfrak{M} & q^2/M^2 + 1 + \frac{1}{2} \bar{g}_2 \mathfrak{N}^2 + \frac{1}{2} \bar{g}_3 \mathfrak{M}^2 \end{pmatrix}. \quad (3.5)$$

If we call α and β the eigenvalues of the mass matrix $\tilde{G}(0)$, we can write for the free energy Γ the result

$$\Gamma[\mathfrak{M}, \mathfrak{N}, t; M, \mu, g_i] = \mathcal{L}_t(\phi = \mathfrak{M}, \chi = \mathfrak{N}) + \frac{\Gamma(\frac{1}{2}\epsilon - 2)}{2(4\pi)^{2-\epsilon/2}} (\alpha^{2-\epsilon/2} + \beta^{2-\epsilon/2}) + \frac{1}{V} \ln C(t), \quad (3.6)$$

where we must include in the Lagrangian counterterms and renormalization constants computed also to the one-loop order. In particular, to the one-loop order, $Z_\phi = Z_\chi = 1$.

The best way to compute $\Delta_2 \Gamma$ is to add yet another term to the Lagrangian $\mathcal{L}_t$ of the form $\sigma \Delta_2$, and compute the new free energy $\Gamma[\mathfrak{M}, \mathfrak{N}, t, \sigma]$ again in the loop-wise expansion, to the one-loop order. Then $\Delta_2 \Gamma$ is obtained as

$$\Delta_2 \Gamma = \frac{\partial}{\partial \sigma} \Gamma[\mathfrak{M}, \mathfrak{N}, t, \sigma] \Big|_{\sigma=0}, \quad (3.7)$$

which yields the result

$$-\Delta_2 \Gamma = \frac{1}{2}(Z_{22}\mathfrak{M}^2 + Z_{21}\mathfrak{M}^2) + \frac{\Gamma(\frac{1}{2}\epsilon - 2)}{2(4\pi)^{2-\epsilon/2}} \left(\alpha^{1-\epsilon/2} \frac{\partial \alpha \sigma}{\partial \sigma} \Big|_{\sigma=0} + \beta^{1-\epsilon/2} \frac{\partial \beta \sigma}{\partial \sigma} \Big|_{\sigma=0} \right), \quad (3.8)$$

where α_σ , β_σ are the eigenvalues of the mass matrix associated to the Lagrangian $\mathcal{L}_t + \sigma \Delta_2$.

B. Equation of state

From the solution of the RGE for $\mathcal{K}$ and $\mathcal{K}_s$ at the critical point, one obtains at once the scaling form of the equation of state. Defining the scaled magnetic fields

$$h = \mathcal{K}(\bar{t}/t)^{2\nu-\epsilon\nu/2}, \quad h_s = \mathcal{K}_s(\bar{t}/t)^{3\nu-\epsilon\nu/2-\eta\nu/2}, \quad (3.9)$$

and the scaled magnetizations

$$m = \mathfrak{M}(\bar{t}/t)^{-2\nu+\epsilon\nu/2}, \quad n = \mathfrak{N}(\bar{t}/t)^{-\nu+\epsilon\nu/2-\eta\nu/2}, \quad (3.10)$$

the equations of state take the scaling form

$$h = f(m, n), \quad h_s = g(m, n). \quad (3.11)$$

Above the critical point the physical condition $h_s = 0$ implies $n = 0$ and the equation of state reduces to a relation between the real magnetic field and magnetization h and m , respectively.

A more accurate solution of the RGE that takes

into account deviations of the effective coupling $g_i(\lambda)$ and $M(\lambda)$ from their fixed-point values and includes the effect of the loop-wise expansion, yields more specific predictions about the equation of state. Instead of looking at the relation between $\mathcal{K}$ and $\mathfrak{M}$ at zero staggered magnetization, which is not very interesting anyway, we shall discuss the relation between the staggered magnetic field and magnetization $\mathcal{K}_s$, $\mathfrak{N}$ at zero real magnetization $\mathfrak{M}$. This equation of state should have the same form as the equation of state of a one-component ferromagnetic system plus corrections due to the coupling with the massive field $\chi(x)$.

If $\mathfrak{M} = 0$ the eigenvalues of the mass matrix $\tilde{G}(0)$ take the simple form

$$\alpha = M^2 \left(1 + \bar{g}_2 \frac{\mathfrak{N}^2}{2} \right); \quad \beta = t + \frac{1}{2} \bar{g}_1^2 \mathfrak{N}^2, \quad (3.12)$$

and it is clear from this result that only β vanishes at the critical point $t = \mathfrak{N} = 0$. The staggered magnetic field is computed by differentiating Eq. (3.6) with respect to $\mathfrak{N}$ at $\mathfrak{M} = 0$,

$$\mathcal{K}_s = t Z_2 \mathfrak{N} + \frac{\bar{g}_1 Z_1}{6} \mathfrak{N}^3 + \delta m^2 \mathfrak{N} + \frac{\Gamma(\frac{1}{2}\epsilon - 1)}{2(4\pi)^{2-\epsilon/2}} \left(\alpha^{1-\epsilon/2} \frac{\partial \alpha}{\partial \mathfrak{N}} + \beta^{1-\epsilon/2} \frac{\partial \beta}{\partial \mathfrak{N}} \right). \quad (3.13)$$

Computing the renormalization constants to order ϵ , and using Eqs. (3.12) for α and β and the solution of the RGE for $\mathfrak{N}_s$, we obtain the result

$$\frac{\mathcal{K}_s}{\mathfrak{N}^6} = \left\{ x + 1 + \frac{1}{6}\epsilon \left[(x+3) \ln(x+3) + x \ln \frac{4}{27} - \ln 27 \right] \right\} [1 + O(t^{\omega\nu})] + c t^{1-2\alpha-\epsilon\nu} (1 + \epsilon \ln t^\nu) x^{\gamma-4\beta}, \quad (3.14)$$

where we have defined, as usual

$$x = t \mathfrak{N}^{-1/\beta}, \quad \omega = \epsilon + O(\epsilon^2), \quad (3.15)$$

and

$$\alpha = \frac{1}{6}\epsilon + O(\epsilon^2); \quad \beta = \frac{1}{2} - \frac{1}{6}\epsilon + O(\epsilon^2); \quad \gamma = 1 + \frac{1}{6}\epsilon + O(\epsilon^2). \quad (3.16)$$

The right-hand side of Eq. (3.14) is identical to the usual result for the equation of state of a one-component ferromagnetic system,³ as expected, except for the last term that represents corrections to the scaling form of the equation of state due to the coupling with the massive field $\chi(x)$.

C. Critical line

We now determine the equation of the critical line that separates the paramagnetic and antiferromagnetic phases near the critical point in the $(\mathfrak{K}, t)$ plane.¹¹ Using the fact that $\Gamma(\mathfrak{M}, \mathfrak{N}, t)$ is the free

energy per unit volume, one can easily show that the critical line $\mathfrak{M} = \mathfrak{M}(t)$ in the $(\mathfrak{M}, t)$ plane is given by the roots of the equation

$$\frac{\partial^2 \Gamma}{\partial \mathfrak{N}^2} \Big|_{\mathfrak{N}=0} = \frac{\partial \mathcal{K}_s}{\partial \mathfrak{N}} \Big|_{\mathfrak{N}=0} = 0, \quad (3.17)$$

because the second derivative of Γ under consideration must be positive in the paramagnetic phase and negative in the antiferromagnetic phase. Moreover, since

$$\frac{\partial \mathcal{K}_s}{\partial \mathfrak{N}} \Big|_{\mathfrak{N}=0} = \frac{1}{\kappa_s} \quad (3.18)$$

κ_s being the staggered susceptibility, it is clear that Eq. (3.10) implies that κ_s diverges on the critical line.

Once again the calculation is done by using the value of Γ given by the loop-wise expansion, Eq.

(3.6), and the appropriate values of the eigenvalues of the mass matrix. By proceeding in the same way as with the calculation of the equation of state, we find

$$-\frac{\partial^2 \Gamma}{\partial \mathfrak{M}^2} \Big|_{\mathfrak{M}=0} = \mu^2 \left(\frac{t}{\bar{t}} \right)^\gamma [A(t)f(t, A(t)) + O(t^{1-\epsilon\nu}) + O(t^2 \ln t)], \quad (3.19)$$

where the function $A(t)$ is of the form

$$A(t) = A - B g_3(\lambda) \mathfrak{M}^2(\lambda); \quad \lambda = (t/\bar{t})^2 \quad (3.20)$$

A and B being some numerical constants. Moreover, the function $f(t, A(t))$ can also be obtained and it turns out to have a finite value at $t=0$. Therefore, the only root of Eq. (3.17) that goes through the critical point $t=\mathfrak{M}=0$ is given to order ϵ by the condition

$$A(t) = 0. \quad (3.21)$$

If we use now the known behavior of the effective coupling constants and magnetizations near the fixed point, we can write $A(t)$ in the form

$$\begin{aligned} A(t) &= A - B g_3 \lambda^{4-\epsilon-1/\nu} \mathfrak{M}^2 \lambda^{-4+\epsilon} \\ &= A - B g_3 (\mathfrak{M}^2/t) \bar{t}, \end{aligned} \quad (3.22)$$

and the roots of Eq. (3.21) are of the form

$$\mathfrak{M}^2 = \text{const } t. \quad (3.23)$$

Since $\mathfrak{M} = \kappa \mathcal{C}$ and, as we will discuss later on, $\kappa(0)$ is finite, Eq. (3.23) implies that the equation of the critical line near the critical point is

$$\mathcal{C}^2 = \text{const } t \quad (3.24)$$

which is in agreement with the predictions of mean-field theories.¹¹ We therefore find no corrections to the equation of the critical line near the critical point. It is interesting to note that this result comes about because of the cancellation that occurs in the behavior of $g_3(\lambda)$ and $\mathfrak{M}^2(\lambda)$ as $\lambda \rightarrow 0$. Indeed, the result follows from the fact that, even in $4-\epsilon$ dimensions, $g_3(t)\mathfrak{M}^2(t)$ behaves like t^{-1} . Had this behavior been different, we would have obtained corrections to the mean-field result for the critical line.

It is remarkable that in most second-order phase transitions that exhibit a critical line the mean-field result is experimentally quite accurate near the critical point. For antiferromagnetic systems, the available experimental data for various substances agree quite well with a linear dependence between $\mathcal{C}^2$ and $T - T_C$ near the Néel temperature.^{11,12} Moreover, the same dependence for small magnetic field $\mathcal{C}$ has been found in the two-dimensional Fisher model¹³ and in numerical calculations carried out by Bienstock¹⁴ for two- and three-dimensional Ising models. Our results are in agreement with these experimental and theoretical indications.

D. Susceptibilities and specific heat

From the expansion (2.29) of the free energy Γ in powers of $\mathfrak{M}$ and $\mathfrak{M}$, one can obtain the susceptibility $\kappa(t)$ and staggered susceptibility $\kappa_s(t)$ in terms of the two-point function at zero magnetic fields and zero momentum,

$$1/\kappa(t) = -\bar{\Gamma}^{(0,2)}(0; t, M, \mu; g_i), \quad (3.25)$$

$$1/\kappa_s(t) = -\bar{\Gamma}^{(2,0)}(0; t, M, \mu; g_i). \quad (3.26)$$

Their behavior as $t \rightarrow 0$ is obtained by solving the RGE for $\bar{\Gamma}^{(0,2)}$ and $\bar{\Gamma}^{(2,0)}$. This can be done both above and below the critical temperature $t=0$, taking into account that for $t < 0$ the field ϕ develops a nonvanishing vacuum expectation value $\mathfrak{M}$.

We find for $\kappa(t)$ the result

$$\kappa(t) = \frac{c_1}{1 - c_2 (t/\bar{t})^{1-\alpha} - c_3 t}. \quad (3.27)$$

This result exhibits a well-known property of $\kappa(t)$, namely, that its maximum is not located at $t=0$ but slightly shifted from the critical point.

The specific heat C can also be obtained by writing it as

$$\begin{aligned} \frac{C}{T_N} &= \frac{\partial^2 T}{\partial t^2} \\ &= \Delta_1^2 \Gamma(0, 0; \mathfrak{M}, \mathfrak{M}, t; M, \mu, g_i), \end{aligned} \quad (3.28)$$

and its critical behavior follows from the RGE for $\Delta_1^2 \Gamma$ with $\mathfrak{M} = \mathfrak{M} = 0$. In this way we obtain a behavior identical to the ferromagnetic case

$$C \simeq \text{const} (t/\bar{t})^{-\alpha}. \quad (3.29)$$

It is interesting to observe that the derivative of $\kappa(t)$, behaves near $t=0$ in the same way as the specific heat. This fact was first established by Fisher from general thermodynamics arguments.¹⁵

As far as the staggered susceptibility is concerned, it behaves in the same way as the susceptibility of a one-component ferromagnetic system, as expected

$$\kappa_s(t) \simeq (t/\bar{t})^{-\gamma}. \quad (3.30)$$

All the results established in this section are the same no matter whether one reaches the critical point $t=0$ from above or from below. This means that $\alpha = \alpha'$ and $\gamma = \gamma'$.

The correlation lengths associated with the real and staggered magnetizations can also be studied near the Néel point. They are given in terms of the two-point functions by

$$\begin{aligned} l_{\mathfrak{M}}^2 &= \Gamma^{(2,0)}(0) / \left. \frac{\partial \Gamma^{(2,0)}}{\partial p^2} \right|_{p^2=0}, \\ l_{\mathfrak{M}}^2 &= \Gamma^{(0,2)}(0) / \left. \frac{\partial \Gamma^{(0,2)}}{\partial q^2} \right|_{q^2=0}, \end{aligned} \quad (3.31)$$

and the result is that l_{3T} diverges at the critical point as $t^{-\nu}$, while l_{3T} is finite and given by

$$\frac{l_{3T}^2(t)}{l_{3T}^2(0)} = \frac{\kappa(t)}{\kappa(0)} \quad (3.32)$$

for temperatures t sufficiently close to the Néel point. The derivative of $l_{3T}^2(t)$ therefore diverges as $t^{-\alpha}$.

Finally, we also want to mention that the subcritical indices that control the next to the leading behavior of all the thermodynamic functions and physical quantities near the critical point discussed in this section, are the same as those of the ferromagnetic case. Corrections to scaling coming from the presence of the massive field are systematically smaller than those already present in the ferromagnetic case.

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